



CycloneShield AI

Cyclone Impact & Infrastructure Vulnerability Forecaster

From **post-landfall recovery** to **pre-landfall action** — asset-level impact forecasts & Gemini-generated early warnings, 24–72 h before landfall.

Build with AI: Code for Communities — Second Edition · **Track 5**

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THE PROBLEM

We forecast the track. We don't forecast what breaks.

- Bay of Bengal cyclones (Fani '19, Amphan '20, Yaas '21, Dana '24) hit Odisha, West Bengal, AP & Bangladesh almost every year.
- Evacuation has improved — but after Fani, Puri & Bhubaneswar were without power for **1-3 weeks**; hospitals, water & telecom failed in cascade.
- Officials receive **where** the storm goes, not **which substation, hospital, road or shelter** fails — and by **when** to act.
- Relief money arrives weeks later; fishers, salt farmers & SHGs sell assets to survive.

1.2 M

people evacuated for Fani
in Odisha

~\$13 B

Amphan losses —
costliest N. Indian Ocean
cyclone

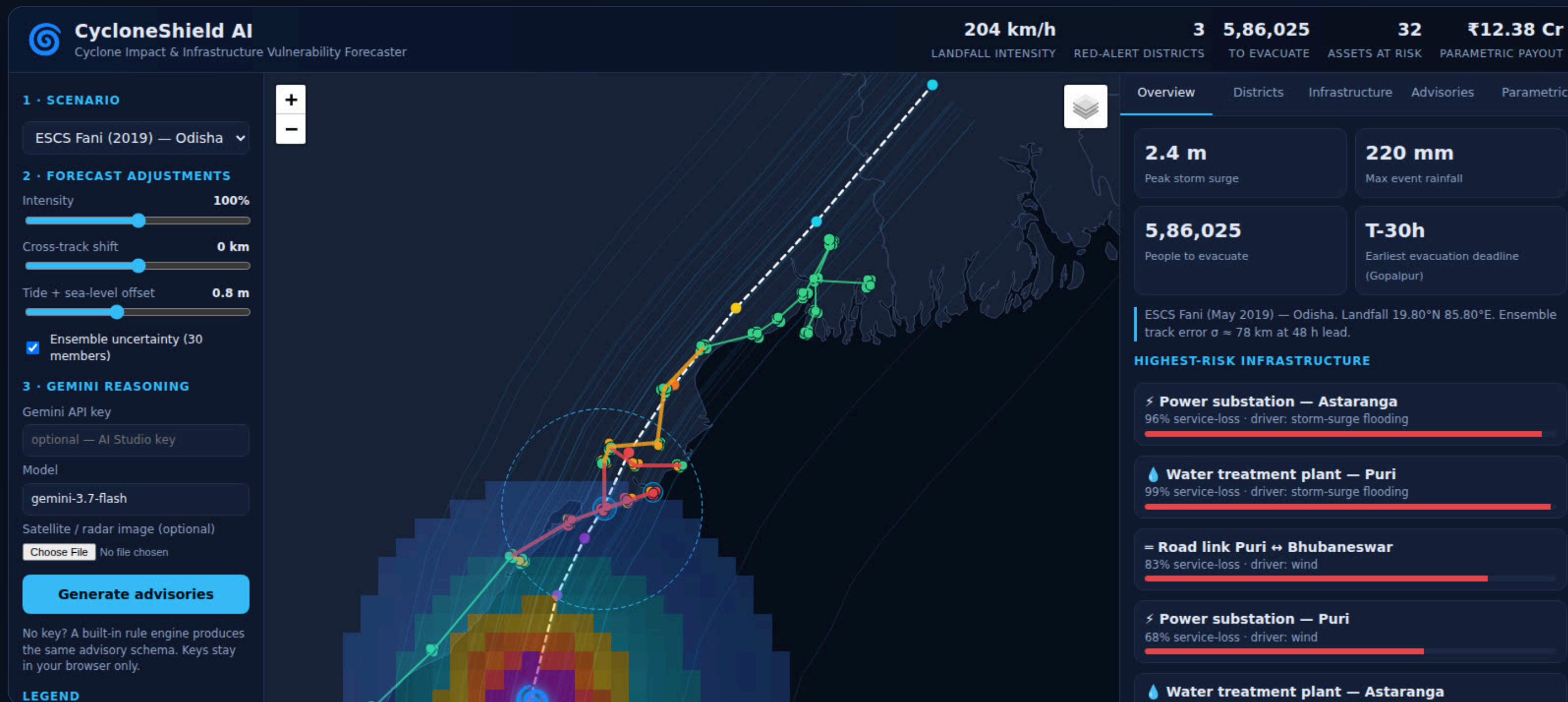
1-3 wk

grid restoration after Fani
in Puri / Bhubaneswar

0

tools linking forecast →
asset failure → action
deadline

Forecast → Impact → Action, in one screen



HOW IT WORKS

Physics-informed hazard engine + Gemini reasoning

1 · Wind field

Holland (1980) parametric profile, B from central pressure, right-of-track asymmetry, inland roughness decay.

2 · Storm surge

Pressure-deficit $\times$ continental-shelf amplification + wind set-up, alongshore decay, tide / SLR, inland attenuation $\rightarrow$ inundation depth.

3 · Rainfall

R-CLIPER-style radial rain-rate integrated hourly along the track $\rightarrow$ event accumulation.

4 · Damage pathways

Lognormal fragility curves per asset (wind / flood / rain) + cascades: grid $\rightarrow$ hospital, water, telecom; road $\rightarrow$ shelter access.

5 · Uncertainty

30-member ensemble (track σ grows with lead time, $\pm 10\%$ intensity) $\rightarrow$ P(damaging wind) per town.

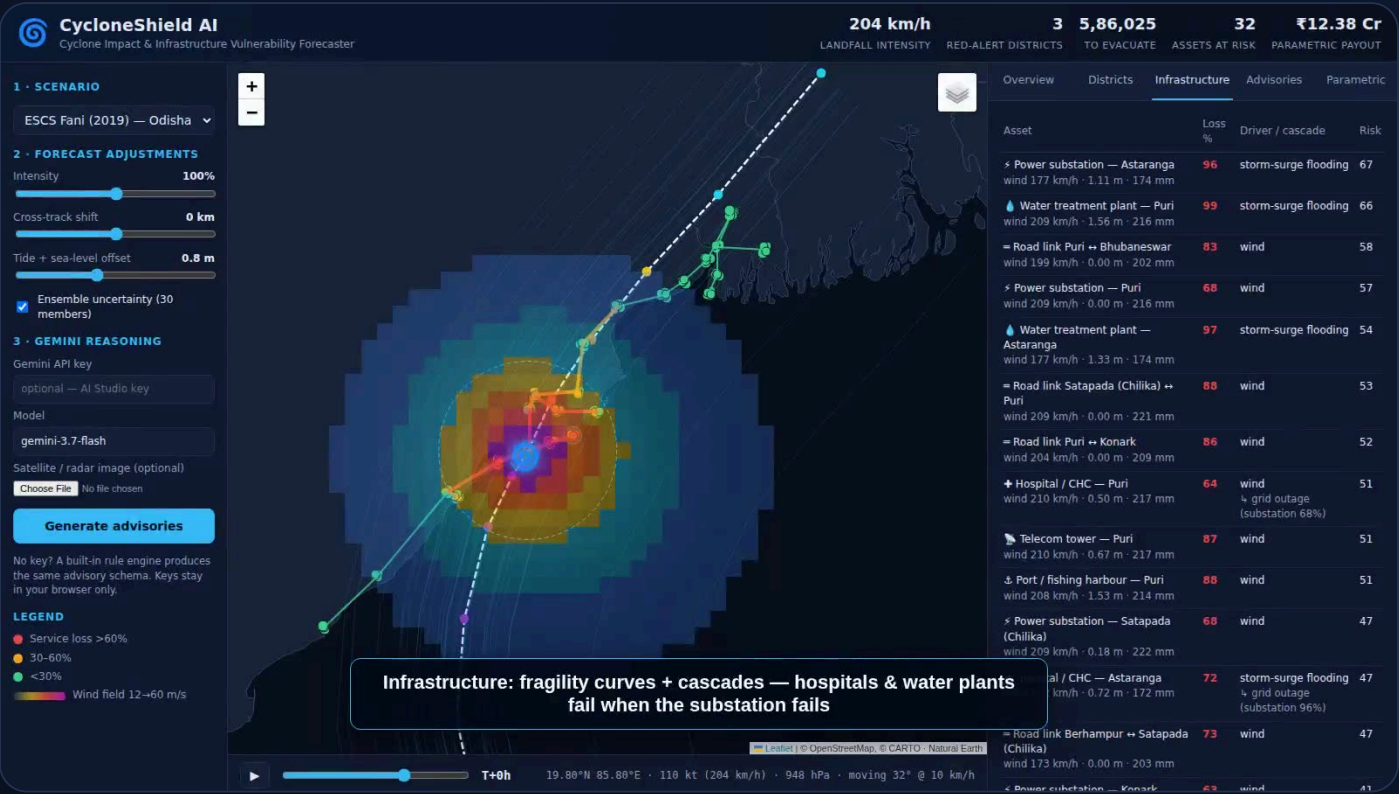
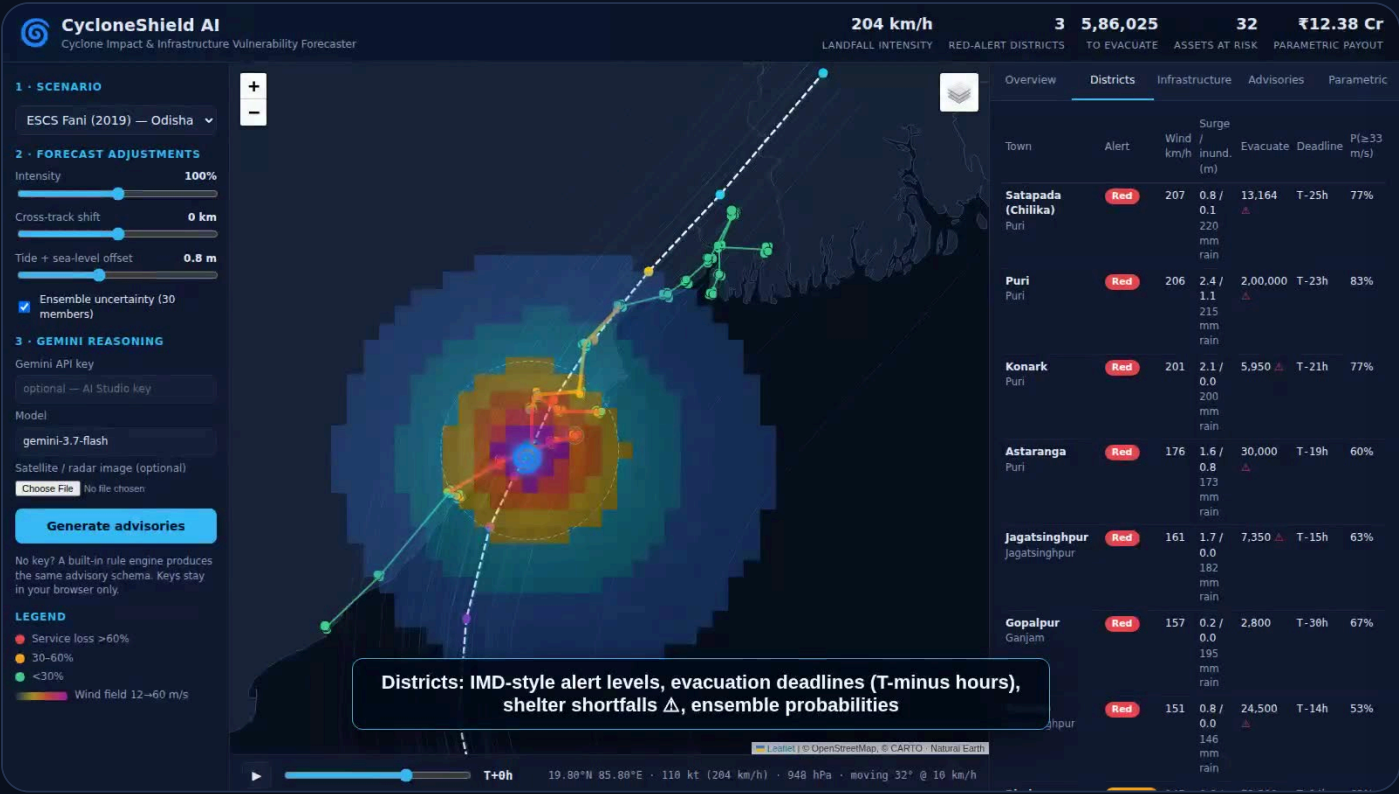
6 · Gemini + dispatch

Multimodal reasoning $\rightarrow$ advisories, native-language SMS, hardening tasks, CAP 1.2 XML dispatch, parametric triggers.

Google Earth Engine supplies exposure: Copernicus DEM low-elevation zones · WorldPop · ESA WorldCover · Sentinel-1 SAR flood validation · GPM IMERG rainfall.

PROTOTYPE

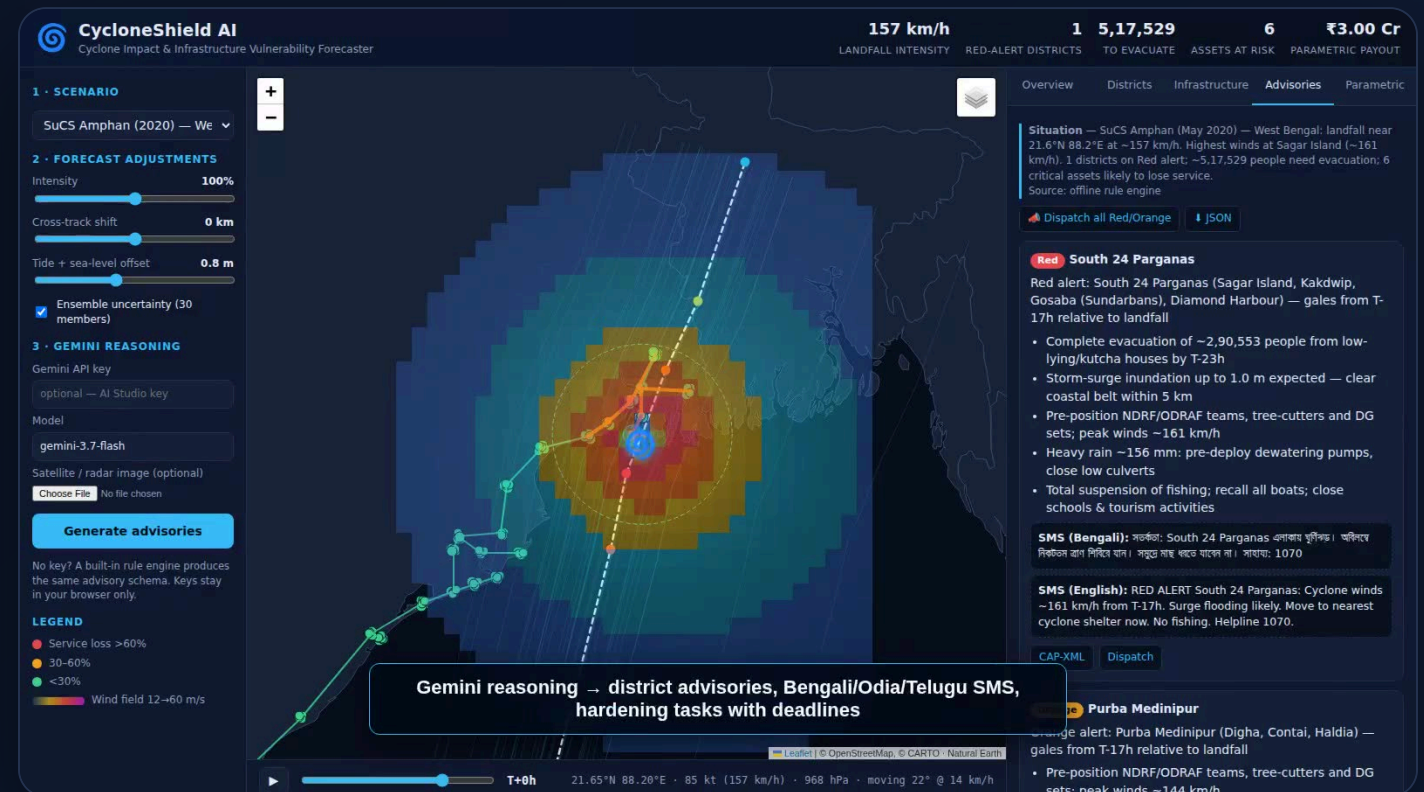
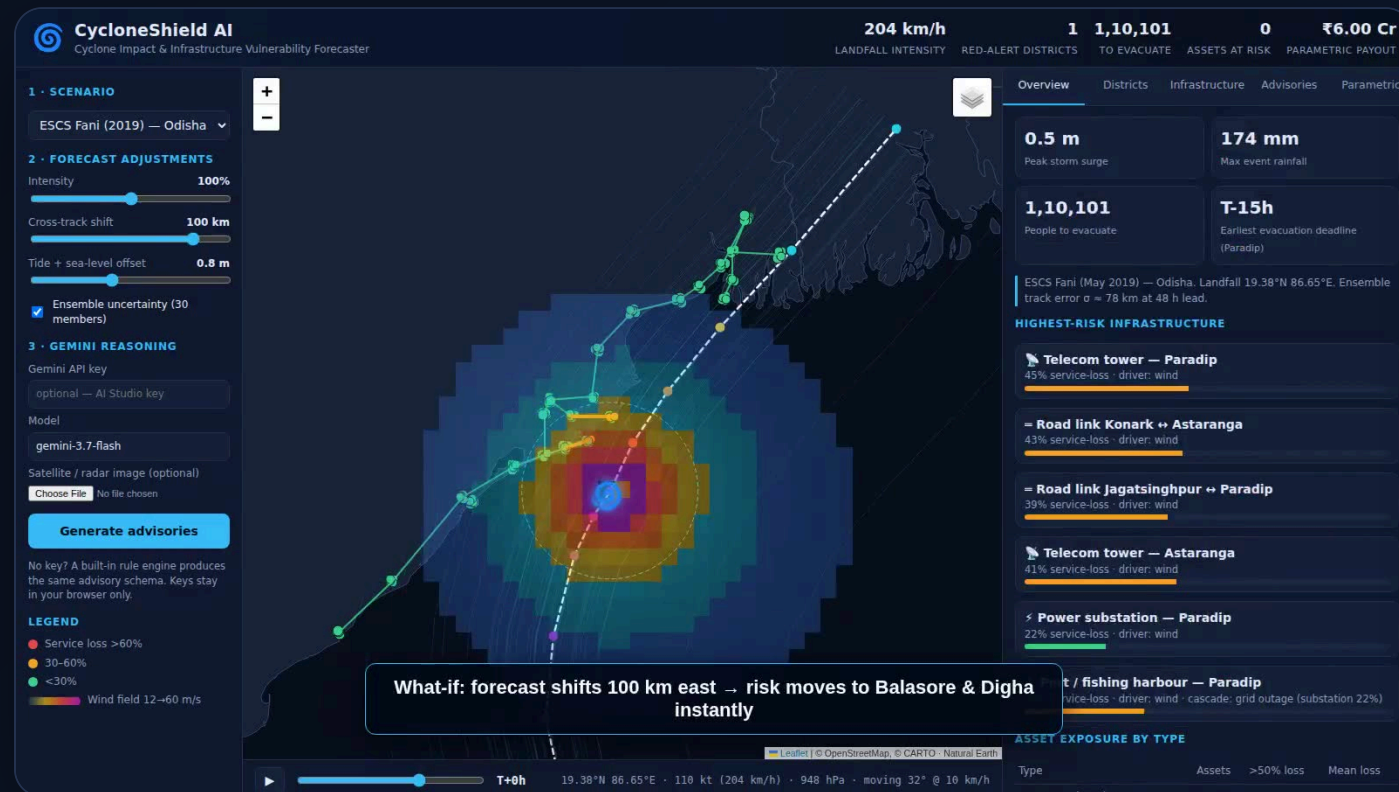
District alerts, deadlines & infrastructure risk



- IMD-colour alerts per town, evacuation deadline = gale onset – 6 h, shelter shortfall warnings Δ .
- Every substation, hospital, shelter, road ranked by service-loss probability, main driver & cascade cause.

PROTOTYPE

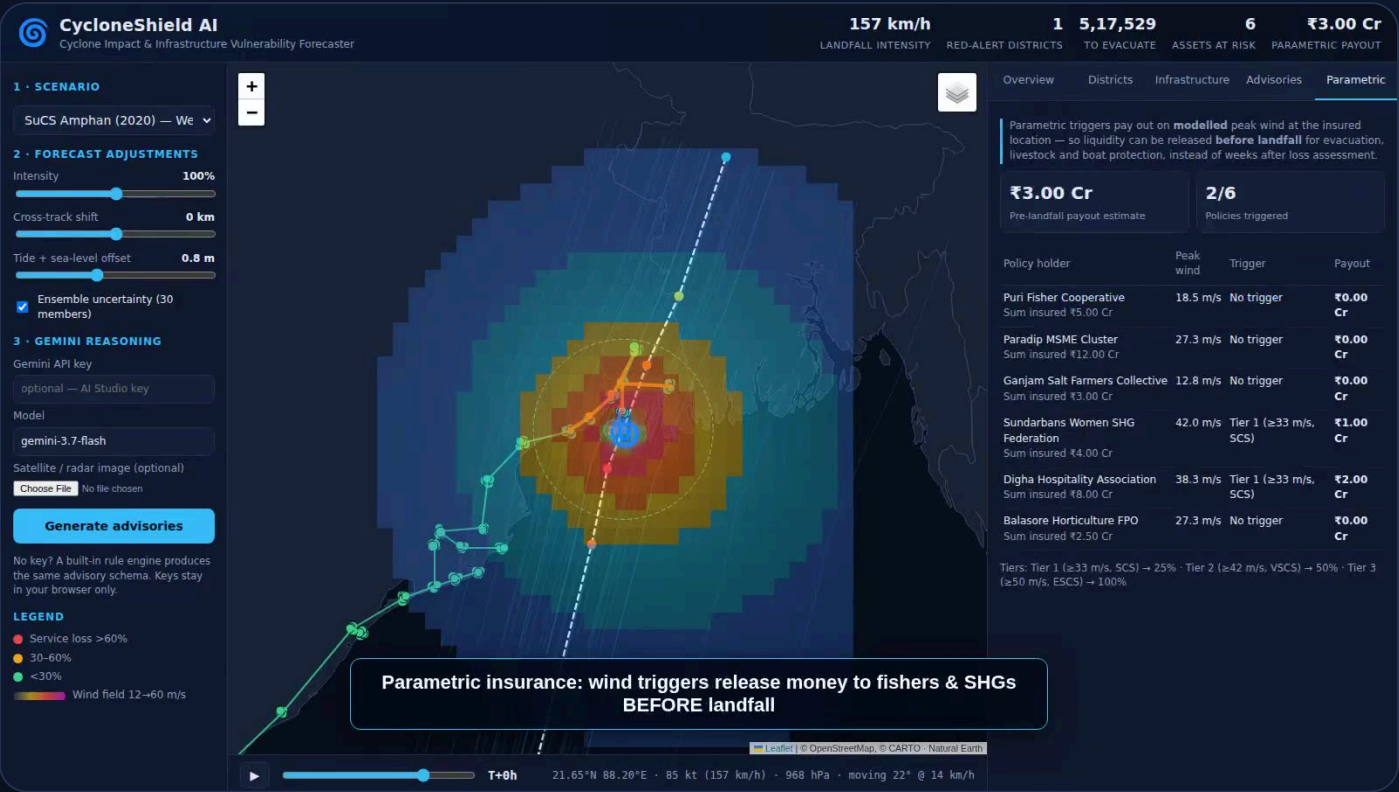
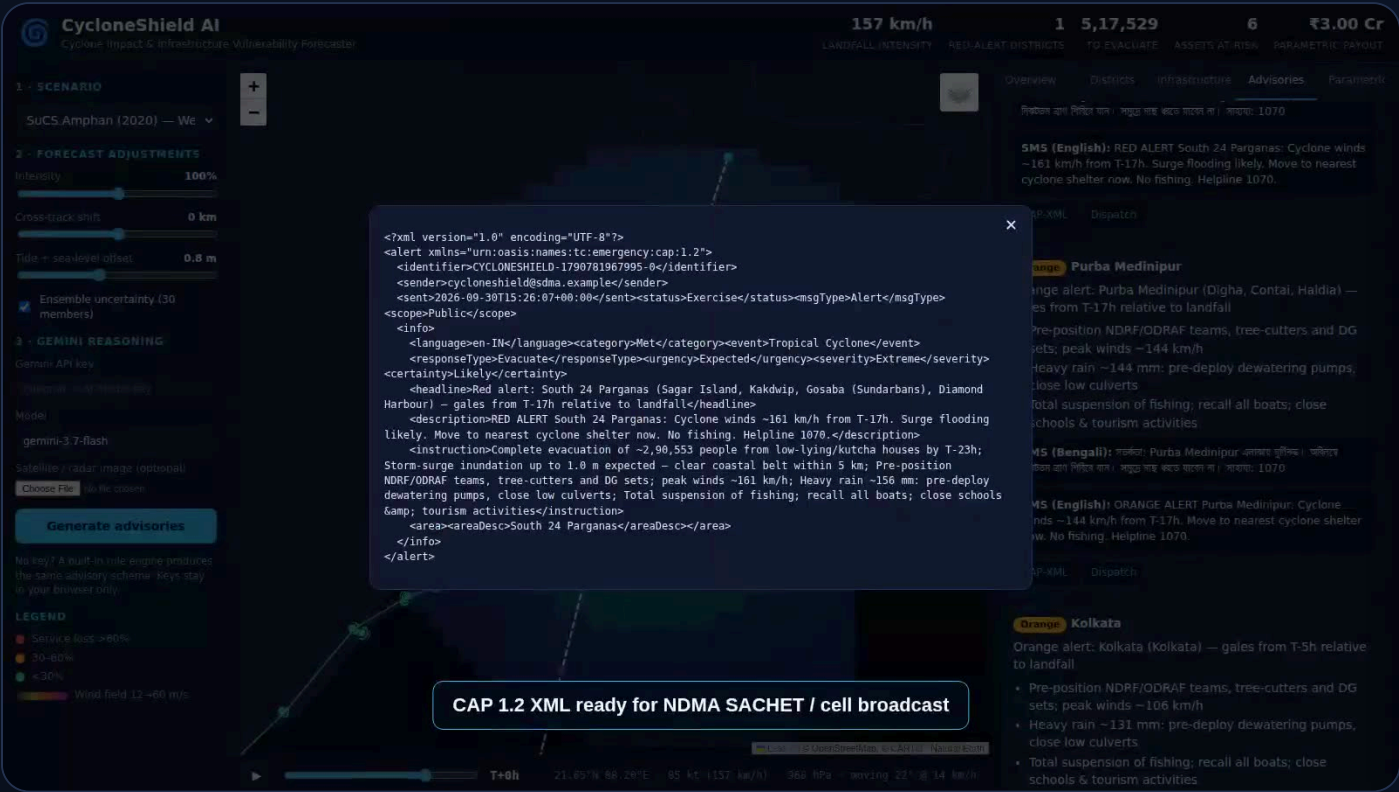
What-if simulation & Gemini advisories



- Shift the track, change intensity or tide — all impacts recompute instantly in the browser.
- Gemini writes district bulletins, Odia / Bengali / Telugu SMS, T-minus hardening tasks; interprets satellite images.

PROTOTYPE

Dispatch in standards & release money early



- **CAP 1.2 XML** — plugs into NDMA SACHET, cell broadcast, SMS gateways & district WhatsApp groups.
- **Parametric insurance** triggers on modelled wind → payouts to fisher co-ops & SHGs **before** landfall.

VALIDATION

Replaying real cyclones

Check	CycloneShield AI	Observed
Fani — peak wind near Puri	~205 km/h	~175–200 km/h
Fani — top failures	Puri / Bhubaneswar substations, grid cascade to hospitals & water	Weeks-long power outage in Puri & Bhubaneswar
Amphan — Kolkata wind	~106 km/h	~112 km/h (Alipore)
Amphan — worst hit	South 24 Parganas / Sundarbans on Red; surge up to ~2.8 m	Sundarbans & South 24 Pgs. most devastated

Honest scope: fast screening models (not ADCIRC); illustrative asset layer anchored on real towns — swap for OSM / utility GIS via the GEE pipeline.

Built for State EOCs, DDMAAs, utilities & insurers

Lives

Town-level evacuation deadlines, shelter gaps, local-language SMS.

Infrastructure

Pre-emptive shutdowns, DG fuel, JCB staging → faster restoration.

Livelihoods

Parametric liquidity for fishers, salt farmers, SHGs, MSMEs.

Scale

Browser-only, offline-capable, CAP-standard; any APAC coast.

Roadmap

- Live IMD / JTWC ingestion on Cloud Run; Firebase auth + FCM for officials.
- OSM + utility GIS assets through Earth Engine; ML surge emulator trained on INCOIS / ADCIRC runs.
- Gemini-assisted post-event Sentinel-1 damage verification & claims automation.



Thank you

Live: ravirepswal108-lgtm.github.io/cycloneshield-ai

Code: github.com/ravirepswal108-lgtm/cycloneshield-ai

Tech: Gemini (multimodal) · Google Earth Engine · Leaflet · Vanilla JS · CAP 1.2